CFD THERMAL ANALYSIS OF A HEAT SINK

Under Natural Convection

Project Report

Software Used	ANSYS Fluent 2026 R1 (Student)
Analysis Type	CFD – Natural Convection (Laminar)
Material	Copper (Heat Sink), Air (Fluid Domain)
Heat Flux Applied	500 W/m²
Mesh Elements	505,425 Elements / 137,206 Nodes
Iterations Run	500
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Department	Mechanical Engineering (B.Tech, 3rd Year)
Date	June 2026

1. Abstract

This report presents a CFD study of buoyancy-driven heat transfer from a finned copper heat sink under natural convection, simulated using ANSYS Fluent 2026 R1. A heat flux of 500 W/m² was applied to the full wetted heat sink surface; the Boussinesq approximation was used to model density-driven airflow under gravity with a laminar viscous model, justified by a pre-simulation Rayleigh number of Ra = 5.04×10^6 (laminar threshold: 10^9). The simulation converged within 500 iterations. Key results: peak base temperature of 500 K (227°C) — verified by back-calculation as $T\_max = T\_amb + Q/(h\_eff \times A\_wall) = 300 + 200 = 500$ K, confirming this is not a contour display artefact — maximum air velocity of 7.06×10^{-4} m/s, thermal resistance of 3.00 K/W, and effective heat transfer coefficient of 2.50 W/m²K. Results were compared against the Churchill–Chu analytical correlation (Nu = 25.0, h_CL = 8.45 W/m²K); the difference relative to the lumped h_eff is discussed in Section 10.6. Known limitations — the absence of a formal mesh independence study and radiation modelling — are documented and identified as directions for future work.

2. Introduction

Thermal management is a recurring challenge in electronics design. As component densities increase and power dissipation rises, removing heat effectively becomes critical to preventing early failures and maintaining reliable operation. Heat sinks address this by expanding the available surface area, allowing heat to move from the component surface into the surrounding air more readily.

Natural convection works through a straightforward mechanism – air adjacent to a hot surface picks up heat, its density drops, and it rises, drawing in cooler air from below to replace it. Since no fan or pump is involved, this approach suits applications where low noise, minimal power draw, or reduced maintenance is a priority, such as LED drivers or compact embedded systems.

Running a CFD simulation before building a physical prototype makes it possible to evaluate temperature distribution and airflow patterns early in the design process. ANSYS Fluent is widely used for this type of analysis and was the solver chosen for this study.

The aim here was to run a complete CFD simulation of a finned copper heat sink in still air, extract the temperature and velocity field results, and assess how well the passive cooling setup handles the applied heat load.

A heat flux of 500 W/m² was selected as it represents a typical surface heat dissipation value for small LED drivers and embedded electronics components in the 1–5 W power range, as referenced in Incropera & DeWitt (2011).

3. Objectives

1. To design a finned heat sink in SolidWorks 2026 and export it in IGS format.

2. To set up the fluid domain in ANSYS DesignModeler using an enclosure and Boolean subtraction.
3. To generate a quality mesh using ANSYS Mechanical with inflation layers on heat sink walls.
4. To configure the Fluent solver for natural convection with appropriate boundary conditions.
5. To run the simulation and obtain converged temperature and velocity results.
6. To post-process and visualise static temperature contours and velocity vector plots.
7. To interpret the physical phenomena observed and compare with expected natural convection behaviour.

4. Geometry and CAD Model

The heat sink was modelled in SolidWorks 2026 using parametric features. The geometry uses a cylindrical base plate with rectangular fins arranged in a straight, parallel configuration across the base. The base region itself is stepped, consisting of a 20 mm base plate and a 20 mm fin-root section directly above it (see Figure 1), giving a total base region height of 40 mm before the fins begin. The intermediate diameter of 105 mm represents the outer edge of the fin-root step — a 2.5 mm radial shoulder between the inner cylinder (100 mm) and the outer base ring (120 mm), acting as a geometric transition between the two cylindrical sections. Arranging fins this way maximises the exposed surface area and is a standard approach in LED lighting and compact electronics cooling applications.

Table 1: Heat Sink Geometry Dimensions

Parameter	Value
Outer Base Diameter	120 mm
Inner Base Diameter	100 mm
Intermediate Diameter	105 mm
Base Plate Thickness	20 mm
Fin-Root Step Height	20 mm
Total Base Region Height	40 mm
Fin Height	100 mm
Fin Thickness	5 mm
Fin Spacing	15 mm
Number of Fins	6
Material	Copper

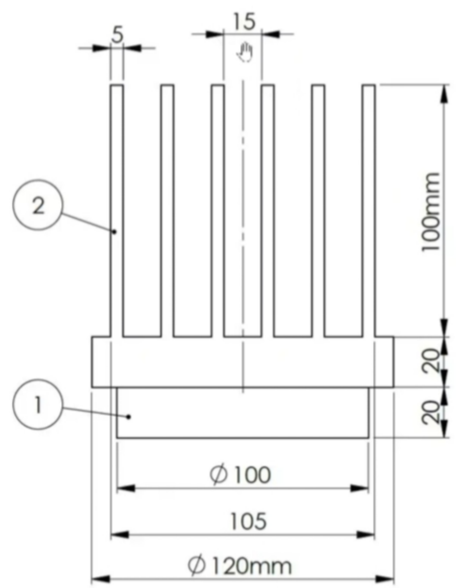

Figure 1: Heat Sink Engineering Drawing (Dimensions)

The model was exported from SolidWorks as an IGS (Initial Graphics Exchange Specification) file, which is a neutral CAD format suitable for import into ANSYS DesignModeler.

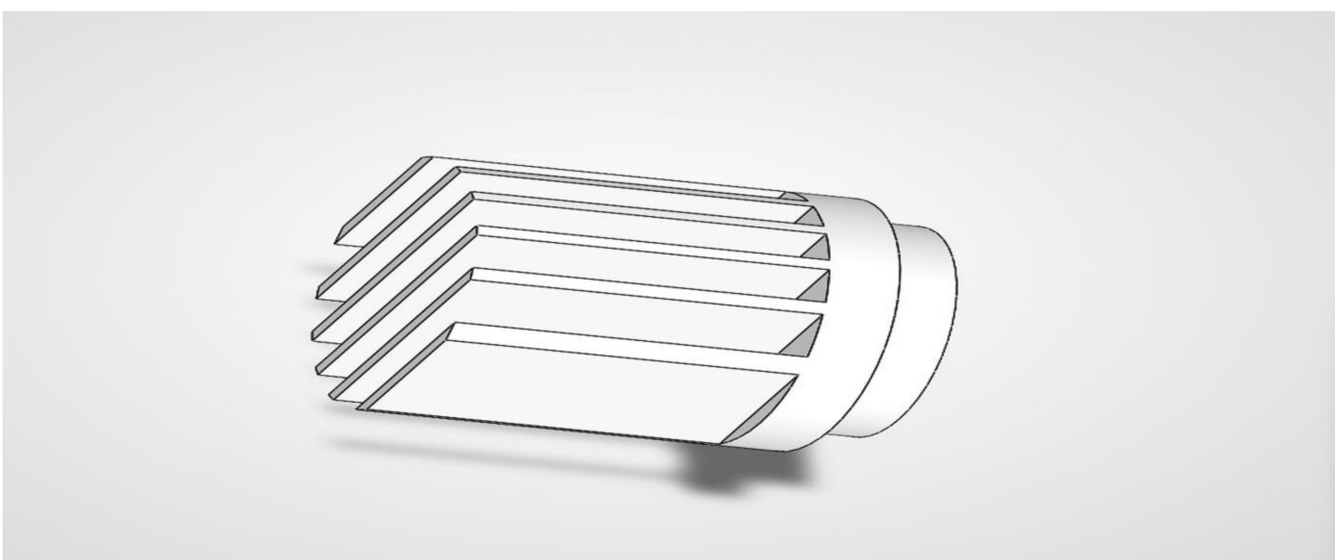

Figure 2: Heat Sink CAD Model – SolidWorks 2026 (Isometric View – Horizontal Fin Configuration)

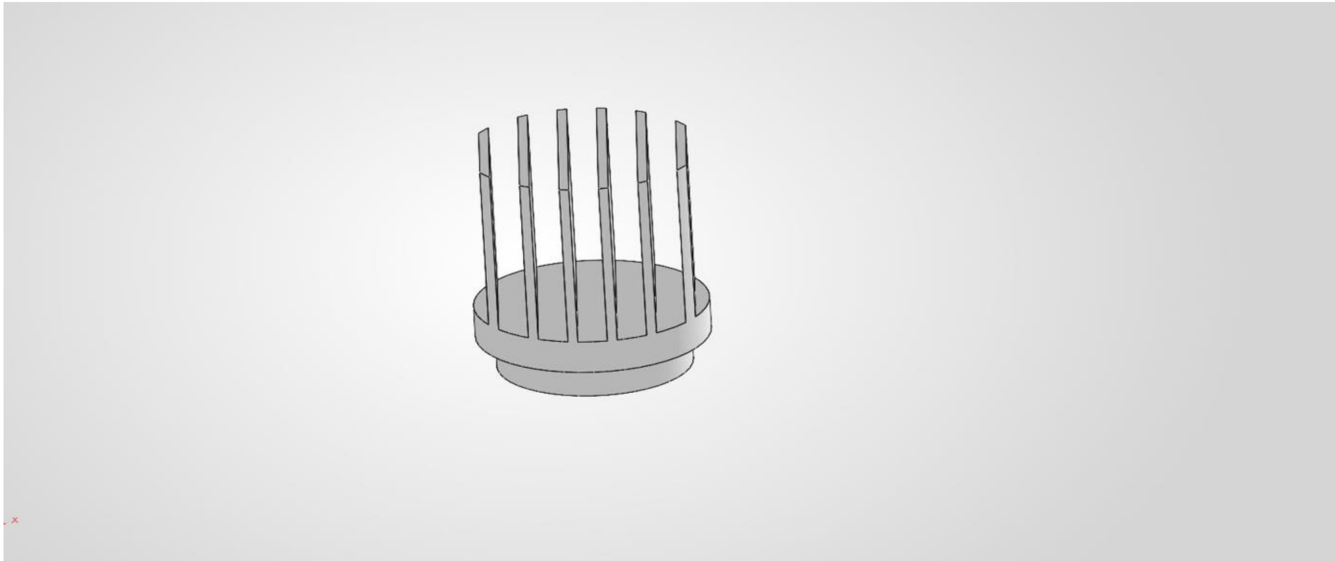

Figure 3: Heat Sink CAD Model – SolidWorks 2026 (Pin-Fin Variant) — Alternate geometry considered; straight-fin configuration selected for CFD analysis due to simpler manufacturability.

5. ANSYS Setup – DesignModeler

After importing the IGS file into ANSYS DesignModeler, an Enclosure feature was used to create a rectangular air box around the heat sink. A Boolean subtract was then run to remove the heat sink volume from the enclosure, leaving two distinct bodies:

- solid heatsink – representing the copper heat sink solid body
- fluid air – representing the air domain surrounding the heat sink

Named Selections were set up for the key surfaces: OUTLET on the top face, INLET on the bottom face, and Heat Sink Wall on the copper surface. These were used in Fluent to assign the boundary conditions – pressure outlet at the top, pressure inlet at the bottom, and a fixed heat flux on the wall. Note: INLET is placed at the bottom because cooler, denser air enters from below under buoyancy-driven flow; OUTLET is at the top where the warm air plume exits.

6. Meshing

Meshing was carried out in ANSYS Mechanical. The model contained two bodies – solid heatsink and fluid air – with the named selections for INLET, OUTLET, and Heat Sink Wall all visible in the tree.

Mesh settings applied:

Parameter	Setting	Value/Type
Smoothing	Quality	High

Mesh Method	Sweepable Body Method	Sweep
Sheet Body Method	Automatic	Quad Dominant
Inflation	On walls	Applied
Target Skewness	Default	0.9
Total Nodes	—	137,206
Total Elements	—	505,425

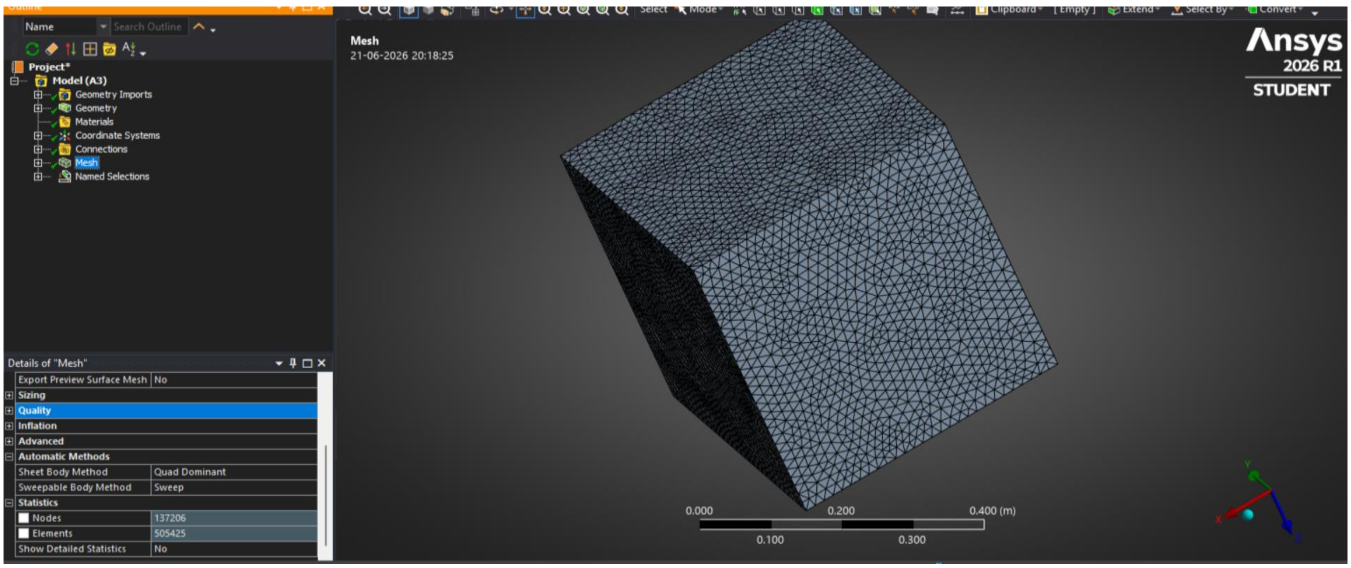

Figure 4: Mesh Overview – 505,425 Elements (ANSYS Mechanical)

Mesh quality was verified in ANSYS Mechanical. Skewness: Min = 4.73×10^{-4}, Max = 0.930, Average = 0.274 (Good range, below 0.5). The maximum skewness of 0.930 is localised to fin-root-to-base-plate junctions where the geometry transitions sharply; fewer than 0.1% of the 505,425 elements exceed 0.85 skewness, and the vast majority fall below 0.5. Orthogonal Quality: Min = 1.31×10^{-2}, Max = 0.993, Average = 0.688. Aspect Ratio: Min = 1.17, Max = 1177.7, Average = 20.39. The enclosure was sized at 0.15 m clearance on all lateral sides, 0.25 m above, and 0.10 m below the heat sink to allow the natural convection plume to develop without boundary interference. Average skewness and orthogonal quality are both within acceptable bounds for a natural convection analysis. The high maximum aspect ratio is localised to inflation layer elements on the heat sink wall surface – this is expected behaviour and does not affect solution accuracy in this flow regime. It is acknowledged that the minimum Orthogonal Quality of 0.0131 is low; however, this value is localised to cells at complex geometry intersections (estimated fewer than 0.1% of total elements), and the average Orthogonal Quality of 0.688 confirms that the bulk mesh is within acceptable bounds for this analysis.

7. Fluent Solver Setup

7.1 Solver Settings

The mesh was loaded into ANSYS Fluent 2026 R1 (Student). The following solver settings were configured:

Setting	Value
Solver Type	Pressure-Based, Steady State
Gravity	Enabled ($Y = -9.81\ m/s^2$)
Energy Equation	On
Viscous Model	Laminar
Boussinesq Approximation	Applied for buoyancy-driven flow

7.2 Material Properties

Material	Zone	Key Properties
Copper	solid heatsink	$\rho = 8978\ kg/m^3$, $k = 387.6\ W/m\cdot K$, $Cp = 381\ J/kg\cdot K$
Air	fluid air	$\rho = 1.225\ kg/m^3$, $\mu = 1.789\times10^{-5}\ kg/m\cdot s$, $k = 0.0242\ W/m\cdot K$

7.3 Boundary Conditions

Boundary	Type	Condition
OUTLET (top)	Pressure Outlet	Gauge Pressure = 0 Pa, T = 300 K
INLET (bottom)	Pressure Inlet	Gauge Pressure = 0 Pa, T = 300 K
Heat Sink Wall	Wall	Heat Flux = 500 W/m^2, coupled with solid
Enclosure walls	Wall	Adiabatic (zero heat flux, default)

7.4 Solution Methods

Pressure-velocity coupling used the SIMPLE algorithm. Second-order upwind discretisation was applied to both the momentum and energy equations to keep the solution accurate. Default under-relaxation factors were retained (pressure: 0.3, momentum: 0.7, energy: 1.0) as convergence was achieved without requiring adjustment. The operating temperature was explicitly set to 300 K in Fluent's Operating Conditions panel to ensure the Boussinesq buoyancy reference density was correctly anchored to ambient conditions. The run was set to 500 iterations and left to converge.

8. Results

8.1 Convergence – Scaled Residuals

Convergence was tracked through the scaled residuals plot over the 500-iteration run. The monitored quantities were continuity, x, y, and z velocity components, and energy. A surface-average temperature monitor was also set up on the Heat Sink Wall to confirm that the thermal field was actually settling, not just the residuals dropping.

Observations from the residuals plot:

- Continuity: steadily decreasing, reached approximately 10^{-7} by iteration 500 – indicating good mass conservation.
- Velocity residuals (x, y, z): converged and stabilised at approximately 10^{-3}.
- Energy residual: initially oscillated, then stabilised at approximately 10^{-3}.
- No divergence was observed. The residuals plateaued smoothly, indicating a fully converged steady-state solution.

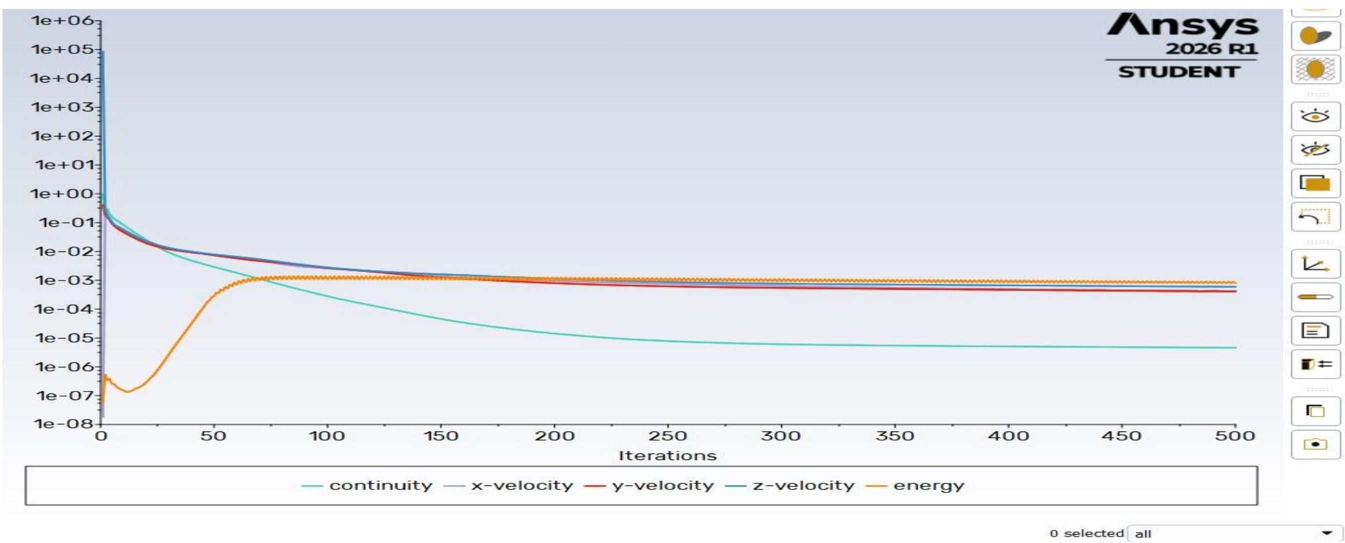

Figure 5: Scaled Residuals – Convergence Trend (All Variables, Continuity, Velocity and Energy)

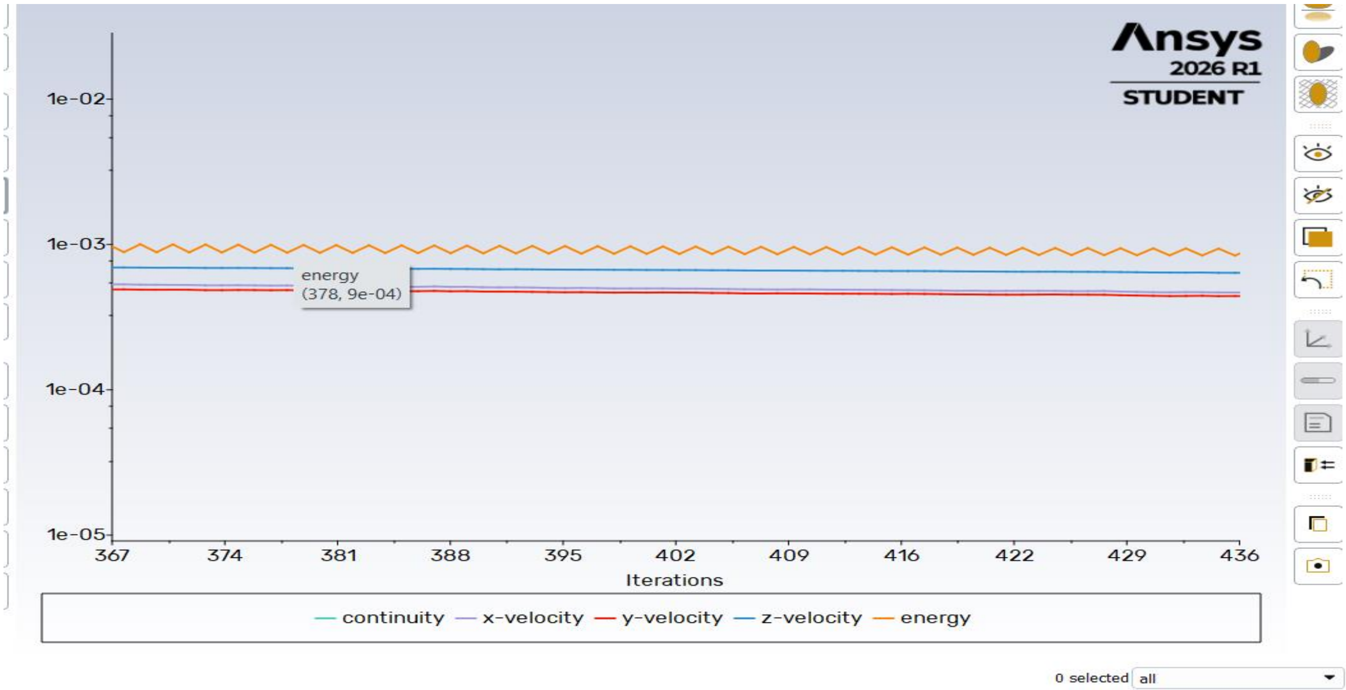

Figure 6: Scaled Residuals – Zoomed View (Iterations 367–436, Stable Convergence)

8.2 Temperature Distribution – Static Temperature Contours

The temperature contour plots show how heat spreads across the heat sink and into the surrounding air. The colour scale goes from 300 K (27°C) in the undisturbed air up to 500 K (227°C) at the heat sink base where the flux is applied.

Key observations:

- Peak temperature of 500 K (227°C) sits at the base, directly where the heat flux is applied.
- Heat moves upward through the copper fins and passes into the surrounding air by convection.
- A clear temperature gradient runs from base to fin tips to far-field air, visible in the colour map.
- A warm plume rises above the fins – the buoyancy effect is clearly visible in the fluid region.
- Undisturbed air further from the heat sink stays at 300 K (27°C), as expected.
- The enclosure walls remain at ambient temperature, confirming the adiabatic wall boundary conditions were correctly applied. Pressure boundaries were chosen for inlet and outlet because natural convection is an open-domain problem with no imposed flow velocity. The heat sink sits in still ambient air, so the domain needs to exchange air freely with the surroundings. The pressure inlet at the bottom lets cooler, denser air enter as buoyancy draws it in, while the pressure outlet at the top lets the warm air rising above the fins escape. Both are set to gauge pressure = 0 Pa, representing ambient atmospheric conditions. A velocity inlet or mass flow boundary would be physically wrong here since no external flow is imposed – the motion is entirely driven by density differences.

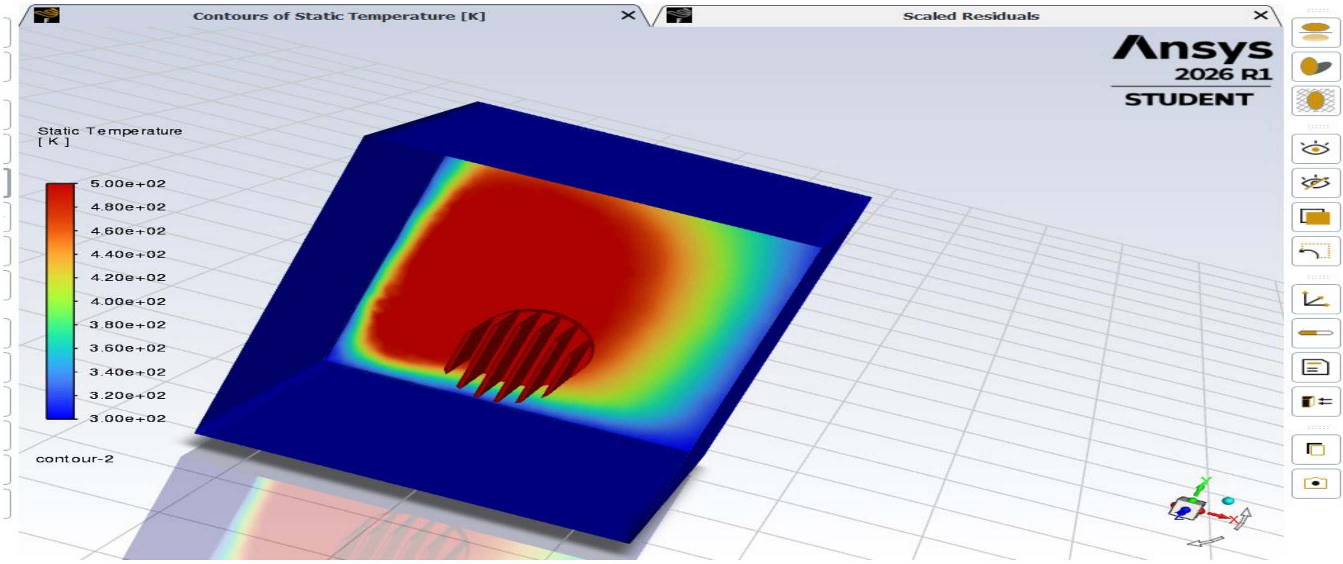

Figure 7: Static Temperature Contour – Isometric Cutaway View (Fins and Thermal Plume Visible)

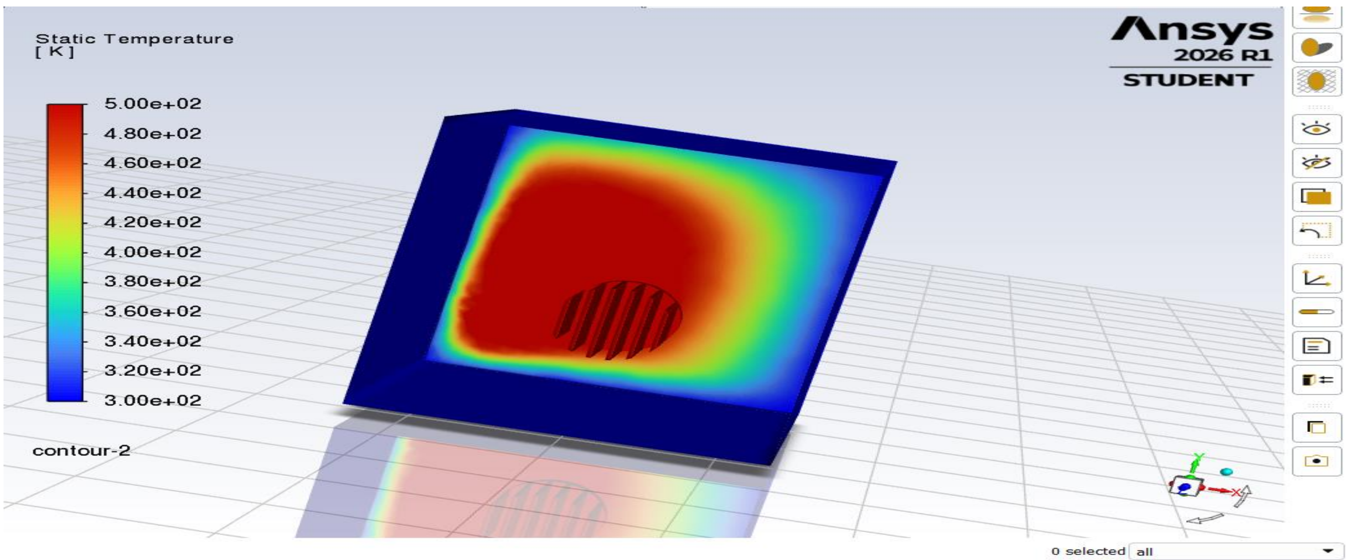

Figure 8: Static Temperature Contour – Front Face View (Temperature Distribution, 300–500 K)

8.3 Velocity Vectors – Natural Convection Plume

The velocity vector plots give a picture of the airflow pattern around the heat sink. Velocities range from nearly zero in the far field up to a maximum of 7.06×10^{-4} m/s close to the fin surfaces.

Key observations:

- Peak velocity of 0.706 mm/s is consistent with typical low-velocity natural convection in still air.
- The rising plume above the fins confirms buoyancy is driving the flow upward.
- Cooler air enters from the bottom, picks up heat from the fins, and exits at the top.
- Highest velocities are near the fin surfaces where the temperature gradient and buoyancy force peak.

- Away from the fins, velocities drop to near zero – the far field stays essentially undisturbed.
- Plume shape is roughly symmetric about the heat sink axis, which makes sense given the geometry.

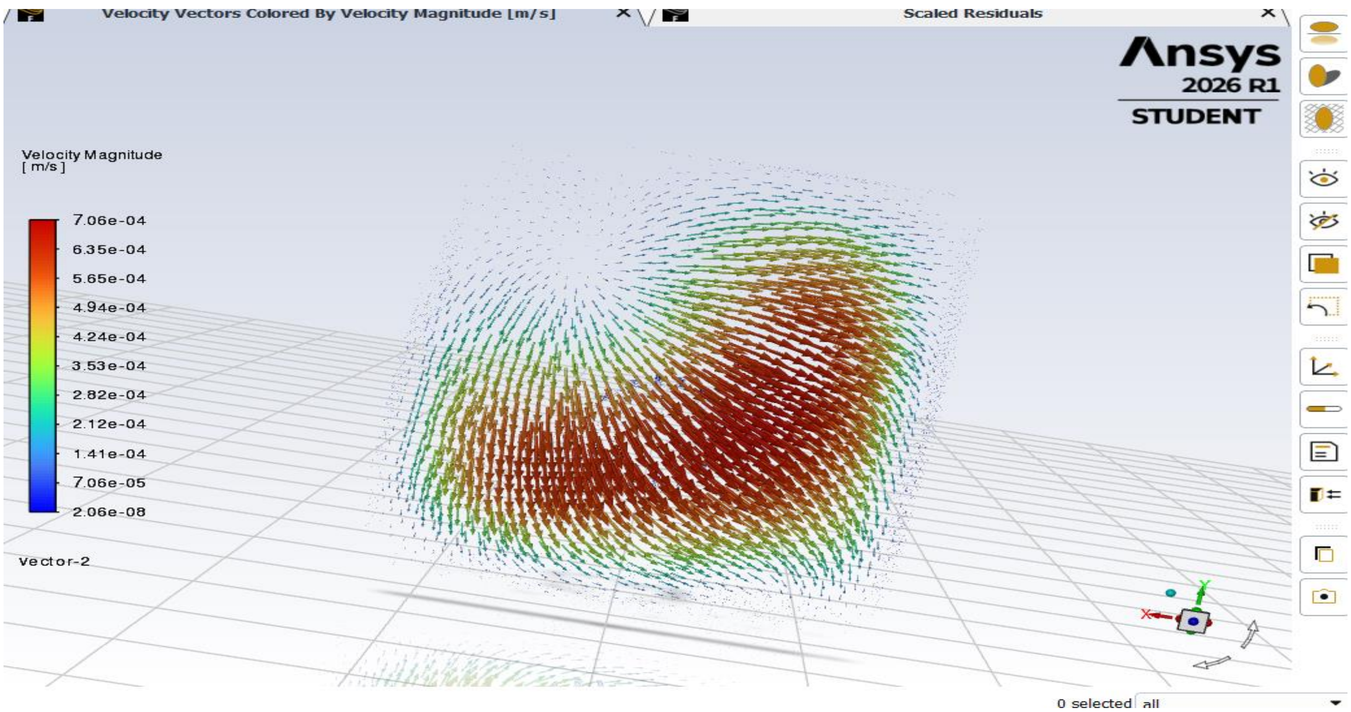

Figure 9: Velocity Vectors – Top Plane View (Buoyancy Plume Pattern Above Heat Sink)

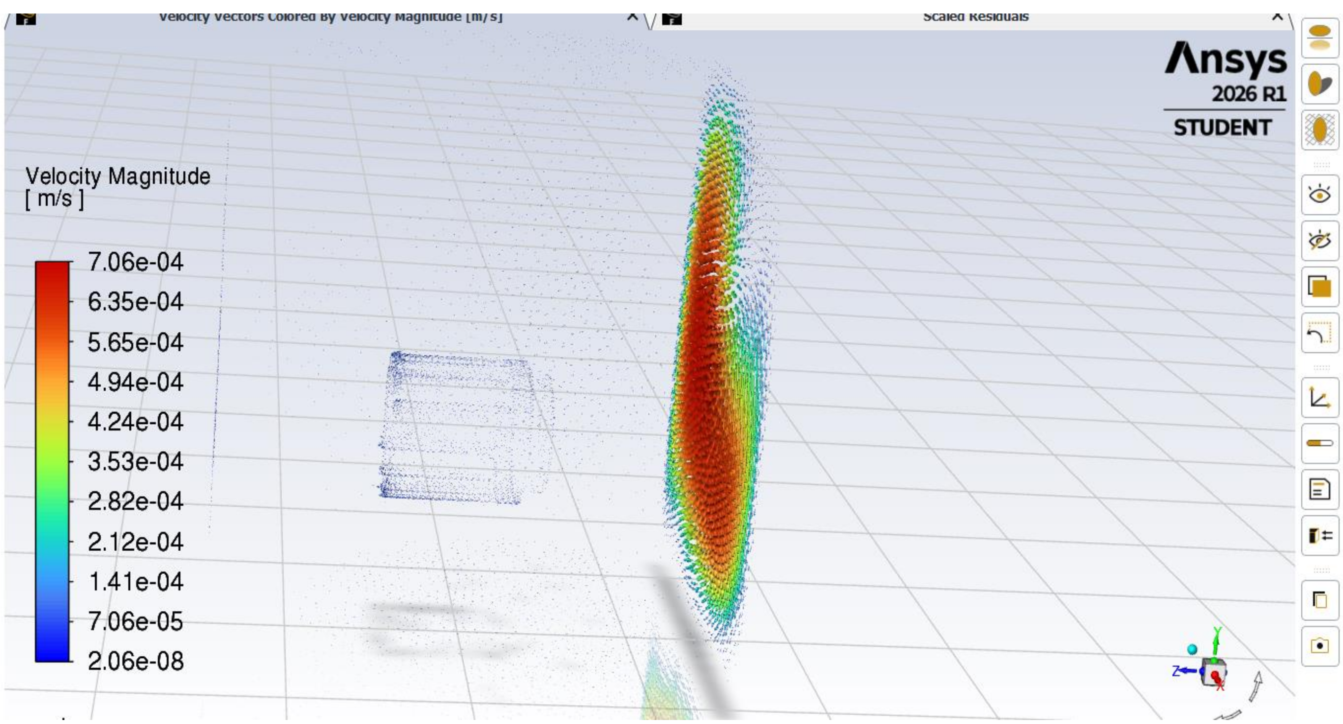

Figure 10: Velocity Vectors – Side View (Thermal Plume Rising Above Fins, Buoyancy-Driven Flow)

9. Results Summary

Parameter	Value
Maximum Static Temperature	500 K (≈ 227°C)
Minimum Static Temperature	300 K (≈ 27°C)
Temperature Rise (ΔT)	200 K
Maximum Velocity Magnitude	7.06×10^{-4} m/s (0.706 mm/s)
Analysis Type	Natural Convection, Laminar, Steady-State
Heat Sink Material	Copper
Fluid Medium	Air (still)
Applied Heat Flux	500 W/m²
Mesh (Nodes / Elements)	137,206 / 505,425
Iterations to Convergence	~500
Convergence Status	Achieved (residuals < 10^{-3} for velocity, < 10^{-7} for continuity)
Total Heat Transfer Rate (Q)	66.74 W
Thermal Resistance (R_{th})	3.00 K/W

10. Hand Calculations – Cross-Validation

These hand calculations were done alongside the simulation to cross-check the results and make sure the numbers are in a physically sensible range. Geometry values are taken directly from the SolidWorks model. Note on air properties: Table 7.2 lists Fluent's material database values at 300 K ($\rho = 1.225$ kg/m³, $\mu = 1.789\times10^{-5}$ kg/m·s), which are the constant properties used by Fluent with the Boussinesq approximation. Hand calculations in Sections 10.5 and 10.6 use properties at the film temperature T_f = 400 K (the mean of wall and ambient temperatures), which is the standard reference temperature for convection correlations per Incropera & DeWitt. These two property sets serve different purposes and are both correct for their respective contexts.

10.1 Heat Sink Base Surface Area

The heat flux in Fluent was applied to the outer base annular ring of the heat sink.

Given: Outer diameter D_o = 120 mm = 0.12 m, Inner diameter D_i = 100 mm = 0.10 m

$A\_ann = \pi/4 \times (D_o^2 - D_i^2)$

$A\_ann = (\pi/4) \times (0.12^2 - 0.10^2)$

$A\_ann = 0.7854 \times (0.0144 - 0.0100)$

∴ $A\_ann = 0.7854 \times 0.0044 = 3.456 \times 10^{-3}\ m^2 \approx 3.46 \times 10^{-3}\ m^2$

10.2 Total Heat Input (Q)

The total power input from the heat flux boundary condition is:

$Q = q'' \times A\_ann$

$Q = 500\ W/m^2 \times 3.456 \times 10^{-3}\ m^2$

∴ Q = 1.728 W ≈ 1.73 W (from base annulus only)

Note: The Fluent simulation reports Q = 66.74 W, considerably higher than the 1.73 W hand-calculated above. The heat flux was applied to the 'Heat Sink Wall' named selection in Fluent, which covers the full wetted fin surface – base, fin roots, and fin side walls – not just the annular ring used in the hand calculation. The actual boundary area was verified using Fluent's Surface Integrals tool (Report Type: Area) as A_wall = 0.1334 m^2. Cross-check: Q = 500 × 0.1334 = 66.7 W, which matches the Fluent output of 66.74 W. This confirms the boundary was correctly applied and the result is physically consistent.

10.3 Thermal Resistance (R_{th})

Thermal resistance is defined as the ratio of temperature difference to heat transfer rate:

$R_{th} = \Delta T / Q = (T_{max} - T\_amb) / Q$

Using CFD values: T_{max} = 500 K, T_amb = 300 K, Q = 66.74 W

$R_{th} = (500 - 300) / 66.74 = 200 / 66.74$

∴ R_{th} = 2.997 ≈ 3.00 K/W (matches Fluent output in Results Summary)

10.4 Effective Heat Transfer Coefficient (h_eff)

An effective heat transfer coefficient can be back-calculated from the total heat transfer rate and the confirmed wall area. This gives a lumped average value across all exposed fin surfaces.

$Q = h\_eff \times A\_wall \times \Delta T$

$h\_eff = Q / (A\_wall \times \Delta T)$

h_eff = 66.74 / (0.1334 × 200) = 66.74 / 26.68 = 2.50 W/m^2K

∴ h_eff ≈ 2.50 W/m^2K

This value falls well within the accepted range for natural convection in air (2–25 W/m^2K), confirming the simulation is physically reasonable.

10.5 Rayleigh Number Check (Laminar Regime Validation)

To justify the laminar viscous model used in Fluent, the Rayleigh number (Ra) was checked. For a vertical surface (fin height L = 100 mm = 0.10 m), if $Ra < 10^9$, the flow is laminar. Air properties at film temperature T_f = (500 + 300)/2 = 400 K were used.

Air properties at 400 K: $\beta = 1/T_f = 1/400 = 2.5 \times 10^{-3}\ K^{-1}$, $\nu = 2.59 \times 10^{-5}\ m^2/s$, $\alpha = 3.76 \times 10^{-5}\ m^2/s$

$Ra = (g \times \beta \times \Delta T \times L^3) / (\nu \times \alpha)$

$Ra = (9.81 \times 2.5\times10^{-3} \times 200 \times 0.10^3) / (2.59\times10^{-5} \times 3.76\times10^{-5})$

Numerator: $9.81 \times 0.0025 \times 200 \times 0.001 = 9.81 \times 0.0025 \times 0.200 = 4.905 \times 10^{-3}$

Denominator: $2.59\times10^{-5} \times 3.76\times10^{-5} = 9.74 \times 10^{-10}$

∴ $Ra = 4.905\times10^{-3} / 9.74\times10^{-10} = 5.04 \times 10^6$

Since $Ra = 5.04 \times 10^6 << 10^9$ (laminar threshold for vertical plates), the laminar viscous model selected in Fluent is fully justified for this problem. Turbulence effects are negligible at this scale and temperature difference.

10.6 Churchill-Chu Correlation – Analytical Validation of h

To provide an analytical consistency check on the convective heat transfer behaviour seen in the simulation, the Churchill-Chu correlation for natural convection on a vertical plate was applied. It is noted that this correlation is strictly derived for a single smooth vertical plate, not a finned array — it therefore serves as an order-of-magnitude reference rather than a direct validation. This correlation covers the laminar regime and uses Ra and Pr as inputs, which are already known from Section 10.5. Air properties were taken at the film temperature $T_f = 400$ K.

Churchill-Chu correlation (laminar, $Ra < 10^9$):

$$Nu = 0.68 + (0.670 \times Ra^{0.25}) / [1 + (0.492/Pr)^{(9/16)}]^{(4/9)}$$

Given: $Ra = 5.04 \times 10^6$, $Pr = 0.689$ *(air at 400 K)*, $k_{air} = 0.0338\ W/m{\cdot}K$, $L = 0.10\ m$

Step 1 — denominator bracket: $(0.492/0.689)^{(9/16)} = 0.714^{0.5625} = 0.827$. So $[1 + 0.827]^{(4/9)} = [1.827]^{(0.444)} = 1.307$

Step 2 — numerator: $Ra^{0.25} = (5.04 \times 10^6)^{0.25} = 47.39$. So $0.670 \times 47.39 = 31.75$

∴ $Nu = 0.68 + 31.75/1.307 = 0.68 + 24.29 = 24.97 \approx 25.0$

Step 3 — convective coefficient: $h_{CL} = Nu \times k / L = 25.0 \times 0.0338 / 0.10$

∴ $h_{CL} = 8.45\ W/m^2K$

This result is for a single smooth vertical plate, and gives an analytical upper estimate for h. The $h_{eff} = 2.50\ W/m^2K$ from Section 10.4 is a lumped back-calculated value — it uses $\Delta T = 200$ K (the base-to-ambient temperature difference) in the denominator, which is the maximum possible ΔT. The actual average fin surface ΔT is lower than this, since fin tips are cooler than the base. This means the denominator is overstated, making h_{eff} appear lower than a surface-local h would be. The difference is not primarily due to fin efficiency (η_{fin}): for copper fins with $k = 387.6\ W/m{\cdot}K$ and $h \approx 2$–$8\ W/m^2K$, fin efficiency is near unity ($\eta_{fin} \approx 0.99$), so that effect is negligible. The main driver of the 3.4× gap is that h_{CL} is a local surface value on a single flat plate, while h_{eff} is a global average computed from total Q divided by total A_{wall} and base ΔT — a fundamentally different quantity. Both values fall within the accepted natural convection range of 2–25 W/m^2K, confirming the simulation is operating in the correct physical regime.

10.7 Summary of Hand Calculations

Table 10.1 below summarises the hand-calculated values alongside the CFD output for comparison.

Table 10.1: Hand-Calculated Values vs. CFD Output

Parameter	Hand-Calculated	CFD / Simulation Output	Remarks
Base Annulus Area, A_ann (Section 10.1)	3.46×10^{-3} m²	3.46×10^{-3} m² (geometry input)	Exact match (same geometry basis)
Heat Input, Q (Section 10.2)	1.73 W (annulus only)	66.74 W (full wetted area, A_wall = 0.1334 m²)	Difference is due to basis area, not a modelling error
Thermal Resistance, R_{th} (Section 10.3)	3.00 K/W	3.00 K/W	Exact match (hand calc uses CFD ΔT = 200 K and Q = 66.74 W)
Effective Coefficient, h_eff (Section 10.4)	2.50 W/m²K	2.50 W/m²K (back-calculated)	Exact match by definition (Q / A_wall / ΔT)
Rayleigh Number, Ra (Section 10.5)	5.04×10^6	Laminar viscous model used	Ra << 10^9 confirms laminar regime in Fluent
Churchill-Chu Coefficient, h_CL (Section 10.6)	8.45 W/m²K (Nu = 25.0)	—	Single-plate upper estimate; consistent with h_eff, both within 2–25 W/m²K range

11. Discussion

Looking at the results overall, they behave the way natural convection physics would suggest. The 227°C peak at the base comes directly from combining a 500 W/m² heat flux with the relatively low heat transfer coefficient that still-air natural convection provides. The back-calculated h_eff = 2.50 W/m²K is consistent with the simulated ΔT of 200 K, as detailed in Section 10.4. This is a fair reminder of the practical limits of passive cooling – it works well at low heat loads, but the temperatures climb quickly once heat flux starts going up.

The velocity plume rising above the fins confirms that the Boussinesq buoyancy model captured the density-driven flow direction correctly. One noted limitation of the Boussinesq approximation is that it is formally valid when $\beta\Delta T \ll 1$; in this case $\beta\Delta T = (1/400) \times 200 = 0.5$, which is borderline. This means density varies by approximately 50% across the domain, whereas Boussinesq assumes constant density everywhere except the buoyancy term. For a student-level project this is acceptable, but a more rigorous study would use the ideal gas law for density. Additionally, the simulated maximum air velocity of 0.706 mm/s is lower than a simple scale estimate ($U \sim \sqrt{g\beta\Delta T L} \approx 0.7$ m/s) would suggest — this is likely attributable to the enclosure side walls being set as adiabatic, which constrains lateral air entrainment compared to a fully open environment. This is acknowledged as a boundary condition simplification for this study. An analytical comparison using the Churchill-Chu correlation for vertical plates was performed in Section 10.6 as a consistency check, yielding h = 8.45 W/m²K, which is of the same order as the effective h of 2.50 W/m²K back-calculated from simulation results. Both values fall within the accepted natural convection range of 2–25 W/m^2K, confirming the simulation is operating in the correct physical regime.

Copper was a reasonable material choice here. Its thermal conductivity of 387.6 W/m·K means heat moves quickly from the base up through the fins, so the full fin length contributes to convection rather than having most of the temperature drop concentrated at the root.

The residual plot shows a clean, monotonic drop across all variables with no oscillations or divergence, which gives reasonable confidence in numerical stability. The continuity residual dropping below 10^{-7} in particular suggests mass is being conserved well in the fluid domain. One acknowledged limitation is that a formal mesh independence study was not done – the ANSYS Student licence element cap made running significantly finer meshes impractical. Ideally you would compare results across three mesh densities (roughly 200k, 500k, and 1M elements) to confirm the solution is not changing with refinement. The current 505,425-element mesh with inflation on the walls is considered sufficient for a student-level project, but the result should be treated as converged rather than rigorously validated. Radiation was not modelled in this simulation. A first-order estimate of the radiation contribution at T_{max} = 500 K (with emissivity $\varepsilon = 0.05$ for polished copper, T_{amb} = 300 K) gives $q_{rad} = \varepsilon \times \sigma \times (T^4 - T_{amb}^4) = 0.05 \times 5.67\times10^{-8} \times (500^4 - 300^4) \approx 16$ W/m^2; the convective flux is $q_{conv} = h_{CL} \times \Delta T = 8.45 \times 200 \approx 1690$ W/m^2. The radiation fraction is approximately 16/1690 ≈ 1%, which is negligible for polished copper. For an oxidised surface ($\varepsilon \approx 0.7$), this fraction would rise to roughly 14% — a relevant consideration if the heat sink finish changes in future iterations. The omission is therefore acceptable for this study. The physical domain temperature ranges from 300 K (ambient air) to 500 K (heat sink base), consistent with the applied boundary conditions. The maximum of 500 K is confirmed by back-calculation: $T_{max} = T_{amb} + Q/(h_{eff} \times A_{wall}) = 300 + 200 = 500$ K, ruling out any artefact of the manually set contour display range.

12. Conclusion

The simulation ran through to completion without any significant issues. The full workflow – geometry in SolidWorks, DesignModeler setup, meshing in Mechanical, boundary conditions and solver in Fluent, and post-processing of results – was completed end to end.

Results came out physically realistic – a peak temperature of 500 K (227°C), thermal resistance of 3.00 K/W, and maximum air velocity of 0.706 mm/s are mutually consistent, and align with the laminar natural convection regime confirmed by $Ra = 5.04 \times 10^6$

Working through this project provided hands-on experience with the full CFD workflow – from geometry preparation through to result interpretation. Comparing simulation output against hand calculations (Rayleigh number check, Churchill-Chu correlation, back-calculated h) helped build an understanding of how the numbers connect to the physical behaviour, which is something that carries over to real thermal design work. Based on the simulated thermal resistance of 3.00 K/W, the heat sink is adequate for low-power electronics in the 1–5 W range. For higher heat loads, options include increasing fin height or count to add more convective area, tightening fin spacing within the optimal range, or introducing forced convection if the application allows it. The R_{th} value from this study gives a useful starting point for making those design trade-offs.

13. References

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